

SHIVANK SAPRA

Minneapolis, Minnesota | [LinkedIn](#) | [Website](#)

EDUCATION

University of Minnesota - Twin Cities

May 2027 (Expected)

Master of Science, Computer Science (4.0 GPA)

Coursework: Machine Learning, Natural Language Processing, Computer Vision, Web Development, Computational Genomics

Guru Gobind Singh Indraprastha University, New Delhi

July 2023

Bachelor of Technology, Computer Science & Engineering

SKILLS

Languages: Python, C++, JavaScript, C#, SQL

AI & Machine Learning: NLP, RAG, Computer Vision, Machine Learning, Deep Learning, Reinforcement Learning

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, Pandas, OpenCV, Matplotlib

Tools & Web Frameworks: Flask, Vue.js, Django, .NET, MVC Architecture, SQL, PostgreSQL, Firebase, Git & Github

PROJECTS

MediRAG: A RAG System for Medical Literature [{Link}](#) | Python, RAG, ChromaDB

- Built an end-to-end RAG pipeline over 200 peer-reviewed PubMed abstracts on mRNA cancer vaccines using Llama3 and ChromaDB.
- Implemented full ingestion pipeline: PubMed E-utilities API, XML parsing, chunking (150-word chunks, 20-word overlap, title prepending) and local vector indexing.
- Designed the retrieval pipeline with sentence-transformers embeddings, cosine similarity search and deduplication to ensure 5 unique cited sources per answer.
- Engineered RAG prompt with source-grounding constraints and low-temperature (0.1) Llama 3 generation to minimise hallucination.
- Evaluated across 30 questions (core domain & off-domain boundary cases) and achieved answer relevancy 0.879, groundedness 0.797 on core domain; and 0.779/0.782 respectively overall.

Linguistic Precursors to Financial Restatements [{Link}](#) | Python, LLM Finetuning, Llama 3

- Built a benchmark dataset linking SEC Form 8-K restatement filings to earnings call transcripts, with a leakage-controlled clean window, company-disjoint splits and SIC/ROA-matched controls.
- Designed a transcript segmentation pipeline to split earnings calls into scripted prepared remarks vs. unscripted analyst Q&A
- Evaluated three model tiers: Loughran-McDonald lexicon baselines, fine-tuned transformers (FinBERT, Longformer) and zero-/few-shot LLM prompting on a pre-restatement financial misconduct detection task.
- Used SHAP attribution to produce interpretable & linguistically grounded explanations of model predictions.

Real-Time Scoring for Rhubarb Dice Game [{Link}](#) | OpenCV, YOLOv8, Computer Vision

- Built a real-time computer vision pipeline to detect & track cards & dice under rotation, occlusion, & motion blur using YOLOv8
- Improved small-object detection by separating detection, dice cropping, color classification, and value classification
- Implemented multi-object tracking (IoU, Kalman, EMA) and stabilized scoring using per-frame state recomputation; evaluated with mAP and precision/recall.

EXPERIENCE

Software Development Engineer | Teleperformance

November 2023 - May 2025

- Developed and maintained scalable frontend and backend systems, including APIs for data-driven applications using C#, .NET Framework, MVC architecture, HTML, CSS, JavaScript.
- Optimized stored procedures in SQL Server by restructuring queries and using indexing strategies, reducing processing times by ~50%.
- Led the development of a collaborative real-time document editor using ReactJS, TypeScript, Express, Socket.io and PostgreSQL, ensuring seamless synchronization.

Software Development Engineer Intern | Teleperformance

December 2021 - April 2022

- Developed a Computer Vision-based Attendance System using Django and OpenCV for seamless attendance management.
- Designed secure role-based access control (Admin/User) to enable personalized dashboards and protected data access.

Machine Learning Intern | GGSIPU USS ACM

August 2021 - September 2021

- Developed a Real-time Drowsiness Detection System with face and eye detection using OpenCV and Haar Cascade Classifiers to enhance road safety by alerting drivers showing signs of fatigue.
- Designed and trained a stacked deep convolutional neural network (CNN) and employed MobileNet for transfer learning to extract dynamic features from keyframes in video sequences. Integrated a Sigmoid classifier layer in the CNN architecture to predict the driver's state as drowsy or alert, triggering an alarm when drowsiness was detected continuously.